

Bachelor Thesis Proposal

Mitigating Time-Shift Errors in CGM-based Glucose Forecasting and Hypoglycemia Event Prediction

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Meta Information

Working title	Mitigating Time-Shift Errors in CGM-based Glucose Forecasting and Hypoglycemia Event Prediction
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1 Motivation and Embedding

Hypoglycemia is a major safety risk in diabetes management. Continuous glucose monitoring (CGM) measures glucose frequently (e.g., every 5 minutes) and enables models that forecast future glucose values or issue early warnings for upcoming hypoglycemia.

In internal forecasting experiments [3], an important practical issue appeared: a model can predict the *right trajectory shape* (rise/fall and general trend) but with a small time offset. For example, the predicted drop may happen 10–15 minutes earlier or later than in the ground truth. Standard pointwise metrics such as RMSE then become large, even though the prediction could still be useful. The same time shift can also affect whether an event warning is counted as correct, because the warning is evaluated at a specific time.

This thesis builds on prior University of Bern work that predicts hypoglycemia events from CGM-only inputs [2]. The main goal is to reduce and better handle time-shift errors, so that (1) forecasting evaluation is less misleading and (2) hypoglycemia warning timing is more reliable.

To keep the scope realistic, this thesis will focus on three concrete directions:

1. **Offset-aware losses:** losses that tolerate small timing offsets during training,
2. **Multi-step forecasting:** training the model to predict the whole future trajectory, not only a single horizon point,
3. **Event-centric alternative:** evaluating (and possibly training) models with a focus on hypoglycemia event detection within a time tolerance window.

The outcome of the thesis will be an experimental comparison of these directions and clear recommendations for training and evaluating CGM-based forecasting and event prediction under small temporal misalignment.

2 Background and Related Work

- **Hypoglycemia event prediction from CGM:** Hüni [2] compares LSTM-based and GAT-based approaches for predicting hypoglycemia events, including recall-focused evaluation (e.g., F_2) [1], [5].
- **Time-shift observations in forecasting:** internal work reports that prediction shifts can dominate RMSE even when the predicted curve is clinically plausible [3].
- **Thesis focus:** instead of proposing a new architecture, this thesis focuses on training objectives and evaluation procedures that explicitly address time shifts.

3 Problem Statement

Let $y_{t:t+H}$ be the true future glucose trajectory over a horizon H , and $\hat{y}_{t:t+H}$ the prediction. Standard evaluation compares values point-by-point, for example with

$$\text{RMSE} = \sqrt{\frac{1}{H} \sum_{k=1}^H (\hat{y}_{t+k} - y_{t+k})^2}.$$

If $\hat{y}$ matches y but is shifted by Δ steps (approximately $\hat{y}_{t+k} \approx y_{t+k+\Delta}$), RMSE can be large although the forecast shape is correct. This effect was observed in internal experiments [3] and can also degrade event timing (e.g., threshold crossing time), which matters for warnings.

Research question. How can CGM forecasting models be trained and evaluated such that small time shifts are reduced and handled explicitly? Concretely, can we reduce the best-alignment shift Δ^* (and its variability across patients and horizons) while maintaining strong hypoglycemia event performance?

Shift measurement. This thesis will quantify time shift using a best-lag value Δ^* computed within a bounded window $\Delta \in [-D, D]$ (e.g., by maximizing cross-correlation). D will be chosen to cover clinically plausible timing offsets.

Clinical relevance. In addition to RMSE/MAE, I will evaluate hypoglycemia event detection using tolerant metrics with a time tolerance τ (a detection counts as correct if it occurs within $\pm\tau$ of the ground truth event time), and report Precision/Recall/ F_1 / F_2 .

4 Objectives and Expected Outcomes

4.1 Objective 1: Offset-aware loss functions

Goal. Reduce the penalty for small temporal misalignment during training, so the model is encouraged to match the trajectory shape and timing more robustly.

Approach. Implement and compare:

- **Bounded-lag alignment loss:** for each training example, evaluate a loss after aligning within $\Delta \in [-D, D]$ (e.g., using correlation-based alignment).
- **DTW-inspired loss (optional):** dynamic time warping (DTW) is a classical alignment method [4]. A DTW-like variant will only be included if it is stable and computationally feasible.

Expected outcome. Compared to pointwise regression loss, offset-aware training should (i) reduce the measured shift Δ^* and (ii) keep event performance at least comparable, ideally improving tolerant event metrics [3].

4.2 Objective 2: Multi-step prediction with intermediate supervision

Goal. Reduce timing drift by training the model to predict intermediate future steps, not only a single horizon output.

Approach. Train models to predict $\hat{y}_{t+1}, \hat{y}_{t+2}, \dots, \hat{y}_{t+H}$ (direct multi-horizon forecasting) or use intermediate supervision in a sequence-decoder setup. The hypothesis is that learning intermediate steps helps the model stay “in phase” over longer horizons [3].

Expected outcome. Multi-step supervision should reduce accumulated timing errors for longer horizons and improve the timing of derived hypoglycemia warnings.

4.3 Objective 3 (Alternative): Focus on event detection instead of exact values

Goal. If perfect alignment remains difficult, prioritize correct and timely event detection over exact pointwise values.

Approach. Evaluate hypoglycemia event detection with a time tolerance τ and report Precision/Recall/ F_1/F_2 (F_2 is recall-focused and consistent with [2]). The thesis will compare:

- events derived from forecasts (threshold crossing on $\hat{y}$),
- direct event prediction (binary labels), consistent with [2].

Direct event training beyond a simple baseline will be treated as optional and only included if time permits.

5 Proposed Methodology

5.1 Data

- **Primary dataset:** the CGM dataset and event labeling approach used in [2] (37 patients, 5-minute sampling).

5.2 Baselines

- **LSTM baseline:** consistent with [2].
- **Patch TST baseline:** consistent with [2].

5.3 Training variants

For each baseline architecture, the thesis proposes:

1. **Standard training:** pointwise regression loss for forecasting or standard classification loss for events.
2. **Offset-aware loss training** (Objective 1).
3. **Multi-step training** (Objective 2).
4. **Event-centric evaluation (and optional training)** (Objective 3).

5.4 Implementation details

- Implementation in Python, using a deep learning framework consistent with prior work (e.g., TensorFlow/Keras as in [2]).
- Patient-wise splits with temporal order to avoid leakage.
- Reproducibility: fixed random seeds and consistent logging of hyperparameters and results.
- Scope control: focus on ablation-style experiments (change one factor at a time), limited hyperparameter tuning.

6 Evaluation Plan

6.1 Metrics

- **Forecasting:** RMSE and MAE.
- **Time shift:** best-lag Δ^* (and distribution statistics such as median and spread across patients), within a bounded window $\pm D$.
- **Events:** Precision, Recall, F_1 , and F_2 , plus tolerant event detection with tolerance τ [2], [3].

6.2 Experimental design

- Patient-wise evaluation using the splitting strategy of [2].
- Compare multiple horizons (e.g., 15/30/60 minutes if applicable).
- Paired comparisons across patients between baseline and each variant.
- Ablations to isolate the effect of each objective.

7 Timeline

The following schedule is organized by phases and key milestones (Weeks 1–20).

Weeks	Phase	Main activities	Key milestones
1–3	Literature review	Read and summarize related work; set up codebase; reproduce baseline and confirm splitting strategy.	M1: Baseline reproduced.
4–6	Protocol and pipeline	Finalize horizons and metrics; implement shift measurement (Δ^* , $\pm D$) and tolerant event evaluation (τ); test on synthetic shifts.	M2: Protocol fixed and pipeline validated.
7–9	Objective 1	Implement offset-aware losses; run baseline vs. Objective 1 experiments; check stability and runtime.	M3: Objective 1 results.
10–12	Objective 2	Implement multi-step forecasting; run controlled experiments under the same protocol.	M4: Objective 2 results.
13–15	Objective 3	Run tolerant event analysis; sweep τ ; compare forecast-derived vs. direct event prediction.	M5: Objective 3 analysis complete.
16–20	Writing	Write methods, experiments, results, discussion; consolidate limitations and recommendations; prepare submission.	M6: Full draft; M7: Submission.

8 Risks, Limitations, and Mitigations

- **Risk: pointwise accuracy and clinical usefulness are not the same.**

Mitigation: report both forecasting metrics and event metrics; explicitly report shift statistics (Δ^*) and use tolerance-based evaluation [3].

- **Risk: class imbalance (hypoglycemia is rare).**

Mitigation: use recall-focused metrics (F_2) as in [2]; report precision–recall trade-offs.

- **Risk: offset-aware losses may be slow or unstable.**

Mitigation: start with a simple bounded-lag method; include DTW-inspired loss only if feasible.

- **Risk: scope creep (too many models/datasets).**

Mitigation: use 1 baseline; treat additional datasets as optional validation only.

- **Risk: limited compute/time.**

Mitigation: limit hyperparameter tuning; focus on ablations and reuse existing pipelines [2], [3].

References

- [1] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
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- [3] Pattern Recognition Group, University of Bern, “Glucose prediction proposal,” Internal unpublished manuscript, no date available.
- [4] H. Sakoe and S. Chiba, “Dynamic programming algorithm optimization for spoken word recognition,” in *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 26, 1978, pp. 43–49.
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